

This Means War

Ahmed Shaikh

"We're going to open a can of whoop ass." Not a line you would expect from the CEO of a multi-million dollar company, but NVIDIA's Chief Executive Officer Jen-Hsun Huang had those fighting words for its now-rival, and processor King—Intel.

Intel and NVIDIA have been exchanging words lately. Intel has the graphics card industry set in sight with its future processor plans. NVIDIA, now the only independent graphic house post the AMD-ATI merger, is not willing to take the threat lying down.

The stakes are high, battle lines have been drawn; information warfare through presentation slides is being fought on an almost daily basis. But what is the battle all about? What brought these seemingly disparate entities—a CPU maker and a GPU maker—to head?

To understand why, we must take a step back in time to 1965, when Intel's Gordon E. Moore put down an observation with far-reaching consequences.

Double The Fun

In the April edition of Electronics Magazine, 1965, Gordon Moore put forth the germ of what

would eventually form the Moore's Law. Moore's original statement read that "The complexity for minimum component costs has increased at a rate of roughly a factor of two per year...", this kernel of an observation was then elaborated by Moore into the statement that is the Law, that the number of transistors on a chip will double about every two years.

Moore's Law can be exploited in two ways, both of which stem from a common root that transistor costs will fall with time. The implications are (1) increase the clock speed of a processor, (2) cram more components onto the surface of a chip.

Until the Pentium 4 processor, increasing the processor speeds was the option to pick. Moore's Law turned into something of a self-fulfilling prophecy for the semiconductor industry. The Law formed the basis for future roadmaps for major silicon companies and thus lent further credence to the law—an electronic ouroboros, eating its own tail. Faster and faster the processors ran, right up to the point where they hit the thermal wall and brought the entire race to a standstill. With the Pentium 4, it was pretty clear that the thermal output of a fast processor took away any positives that a fast chip might bring to board.

It was time for a rethink.

A look at Intel's upcoming GPU rival—the Larrabee processor—we reveal the why, the what, and the so-what?

